



Physiology of Metabolism

Part One

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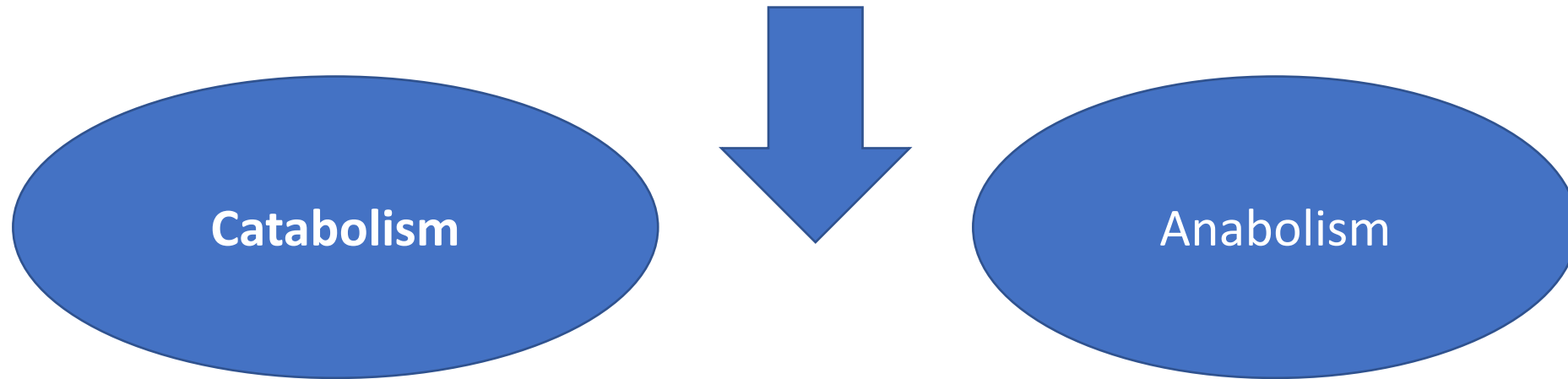
Learning Outcomes:

- Define metabolism.
- Explain the role of ATP in anabolism and catabolism.
- Describe the fate and metabolism of carbohydrates.

What is Metabolism?

- **Metabolism** refers to all of the chemical reactions that occur in the body.

There are two types of metabolism



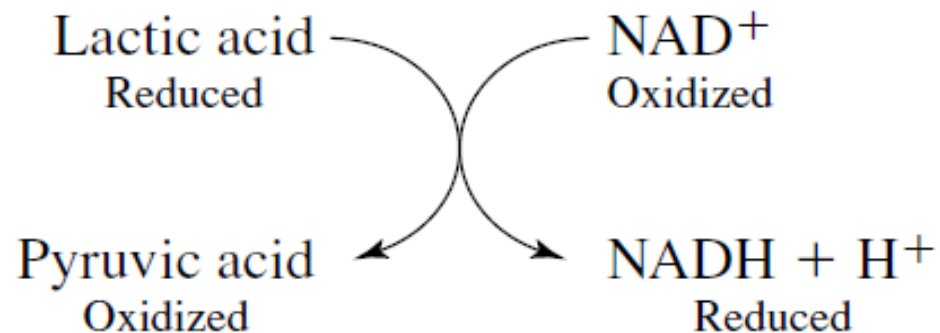
Catabolism:

- Those chemical reactions that break down complex organic molecules into simpler ones are collectively known as catabolism.
- Overall, catabolic (decomposition) reactions are **exergonic**.
- **Exergonic Reactions**  produce more energy than they consume, releasing the chemical energy stored in organic molecules.
- Important sets of catabolic reactions occur in glycolysis, the Krebs cycle, and the electron transport chain

Anabolism:

- Chemical reactions that combine simple molecules and monomers to form the body's complex structural and functional component are collectively known as anabolism.
- Anabolic reactions are **endergonic**.
- **Endergonic Reactions**  they consume more energy than they produce.
- Examples of anabolic reactions are the formation of peptide bonds between amino acids during protein synthesis, the building of fatty acids into phospholipids that form the plasma membrane bilayer, and the linkage of glucose monomers to form glycogen.

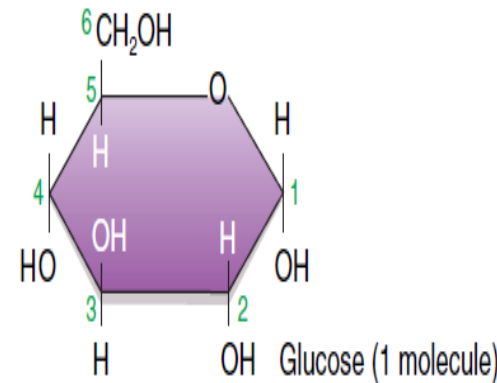
- When a substance is oxidized, the liberated hydrogen atoms do not remain free in the cell but are transferred immediately by coenzymes to another compound.
- Two coenzymes are commonly used by cells to carry hydrogen atoms: **nicotinamide adenine dinucleotide (NAD)**, and **flavin adenine dinucleotide (FAD)**.



- e.g: when a cell oxidizes a molecule of **glucose (C₆H₁₂O₆)**, the energy in the glucose molecule is removed in a stepwise manner. Ultimately, some of the energy is captured by transferring it to ATP, which then serves as an energy source for energy-requiring reactions within the cell.
- Compounds with many hydrogen atoms such as glucose contain more chemical potential energy than oxidized compounds. For this reason, **glucose** is a valuable nutrient.

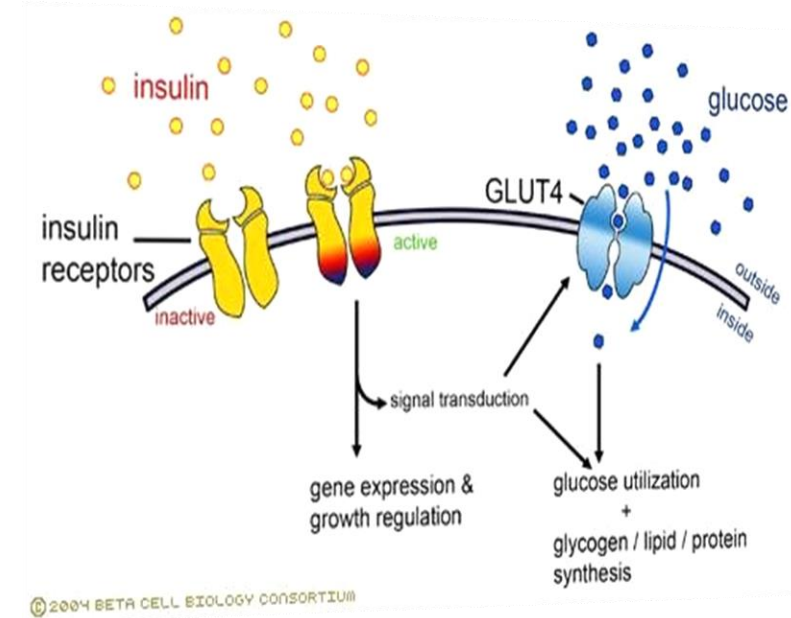
Carbohydrate Metabolism

- Both polysaccharides and disaccharides are hydrolyzed into the monosaccharides glucose (about 80%), fructose, and galactose during the digestion of carbohydrates.
- Some fructose is converted into glucose as it is absorbed through the intestinal epithelial cells.
- Hepatocytes (liver cells) convert most of the remaining fructose and practically all of the galactose to glucose.
- *So the story of carbohydrate metabolism is really the story of glucose metabolism.*



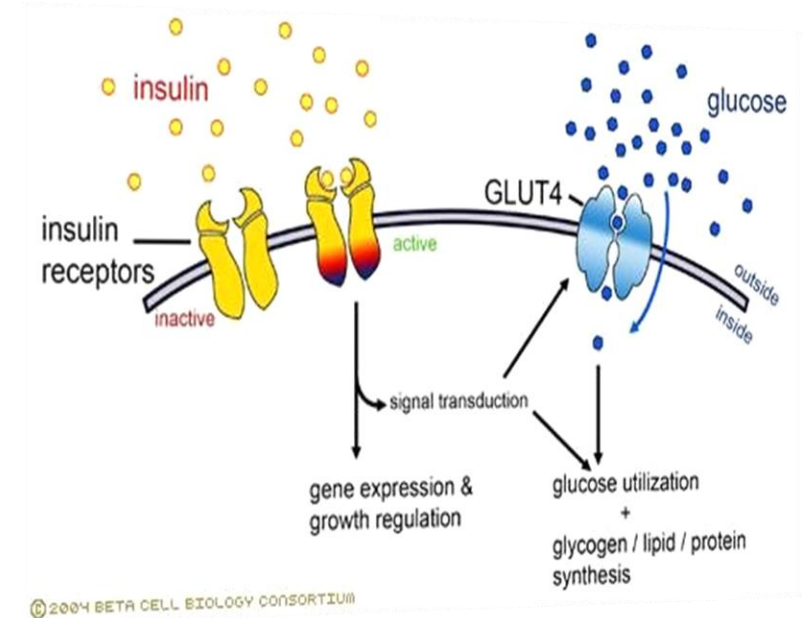
Glucose Movement into Cells:

- Before glucose can be used by body cells, it must first pass through the plasma membrane and enter the cytosol.
- Glucose absorption in the gastrointestinal tract (and kidney tubules) is accomplished via **secondary active transport (Na–glucose symporters)**.
- Glucose entry into most other body cells occurs via **GLUT molecules**, a family of transporters that bring glucose into cells via **facilitated diffusion**.



Glucose Movement into Cells:

- A high level of insulin increases the insertion of one type of GluT, called **GLUT4**, into the plasma membranes so increasing the rate of facilitated diffusion.
- In neurons and hepatocytes, another type of GluT is always present in the plasma membrane, so glucose entry is always “turned on.”



Glucose Catabolism:

The oxidation of glucose to produce ATP is also known as cellular respiration, and it involves 4 sets of reactions:

1) Glycolysis: two molecules of **pyruvic acid**, two molecules of ATP and two energy-containing NADH H^+ are produced.

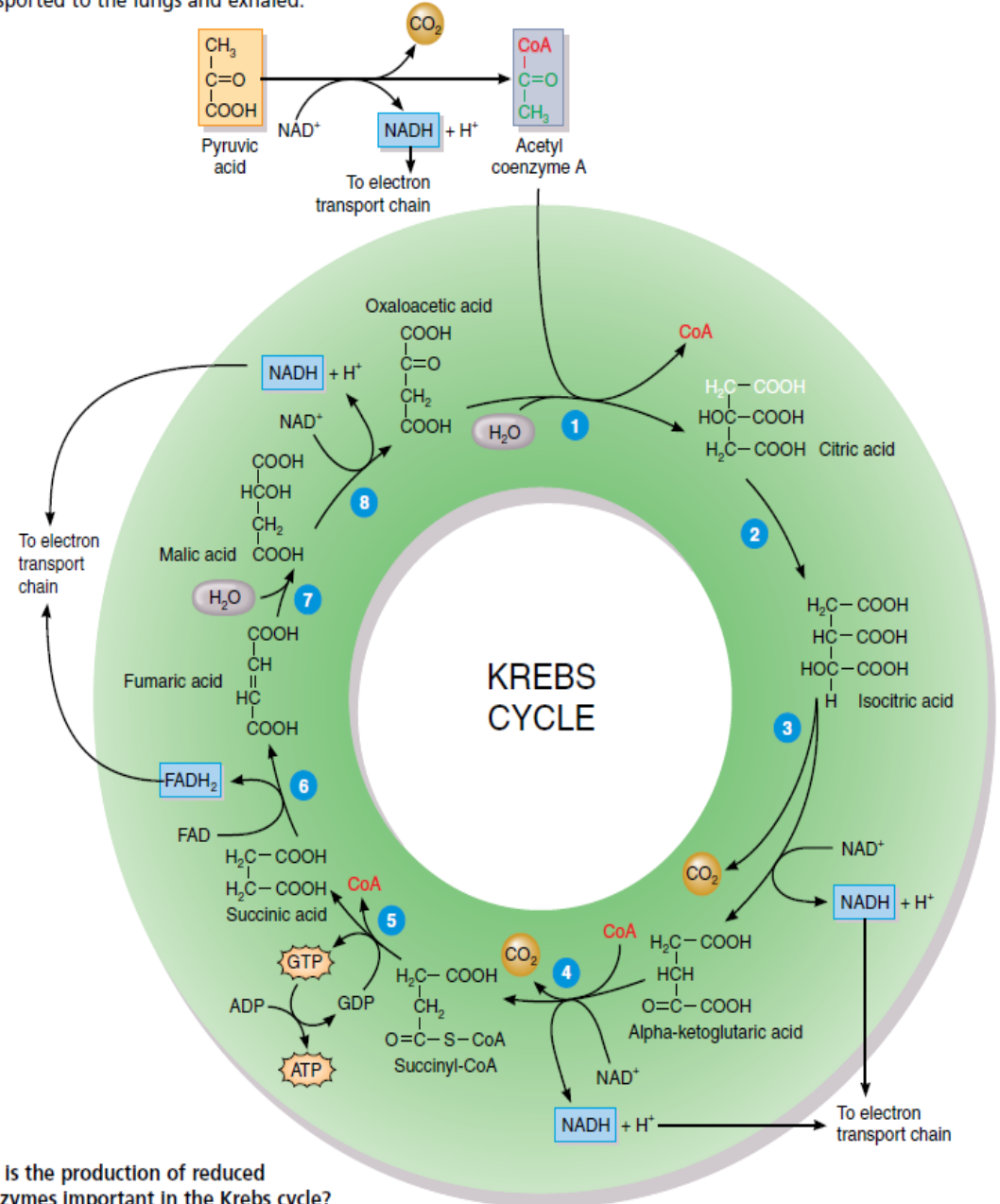
2) Formation of acetyl coenzyme A: A transition step that prepares pyruvic acid for entrance into the Krebs cycle. This step also produces energy-containing NADH H^+

Glucose Catabolism:

3) Krebs cycle reactions: reactions oxidize acetyl coenzyme A. ATP, NADH H, and FADH₂ are produced.

4)Electron transport chain (ETC) reactions: These reactions oxidize NADH H and FADH₂ and transfer their electrons through a series of electron carriers.

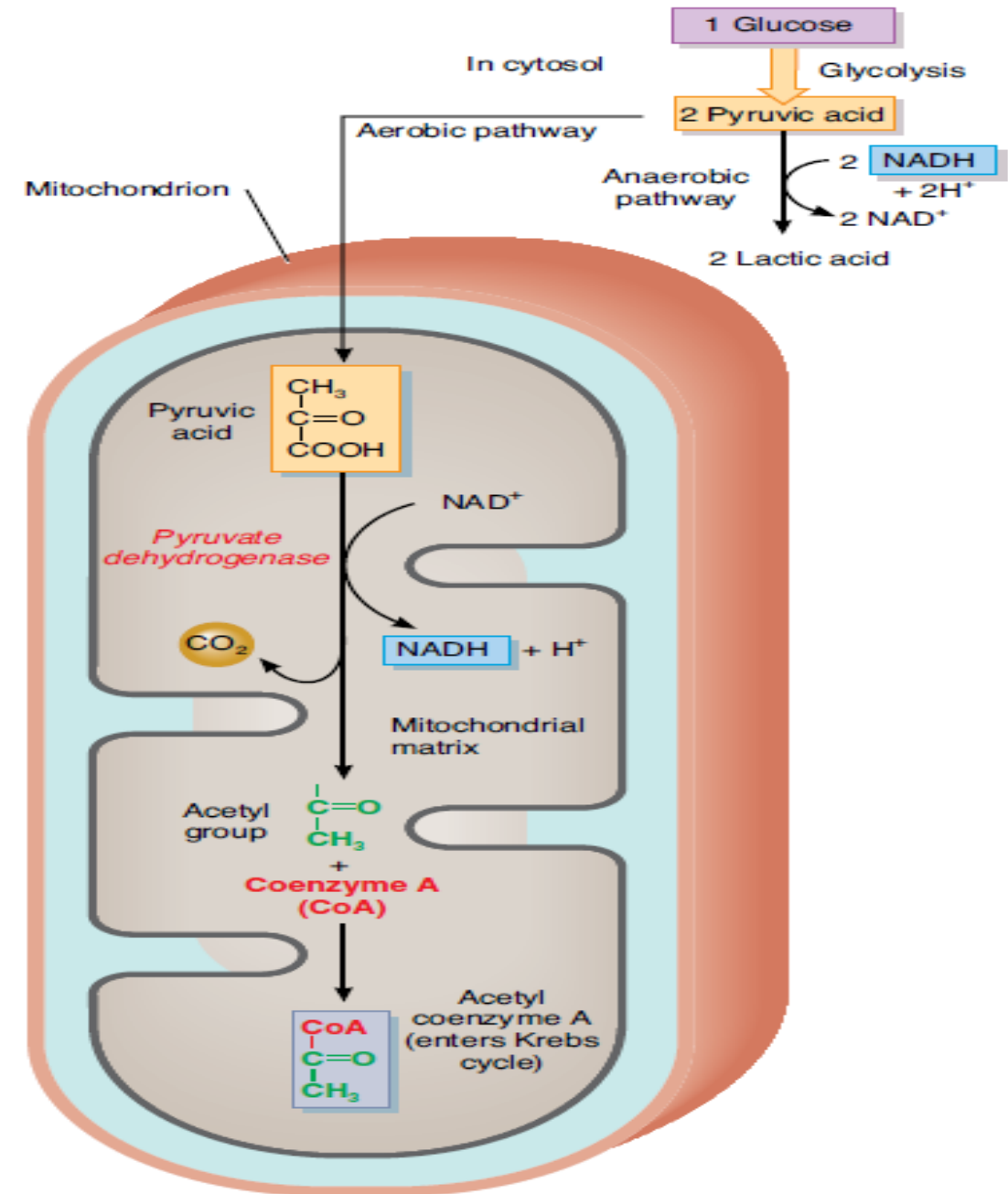
Key: The three main results of the Krebs cycle are the production of reduced coenzymes (NADH and FADH₂), which contain stored energy; the generation of GTP, a high-energy compound that is used to produce ATP; and the formation of CO₂, which is transported to the lungs and exhaled.



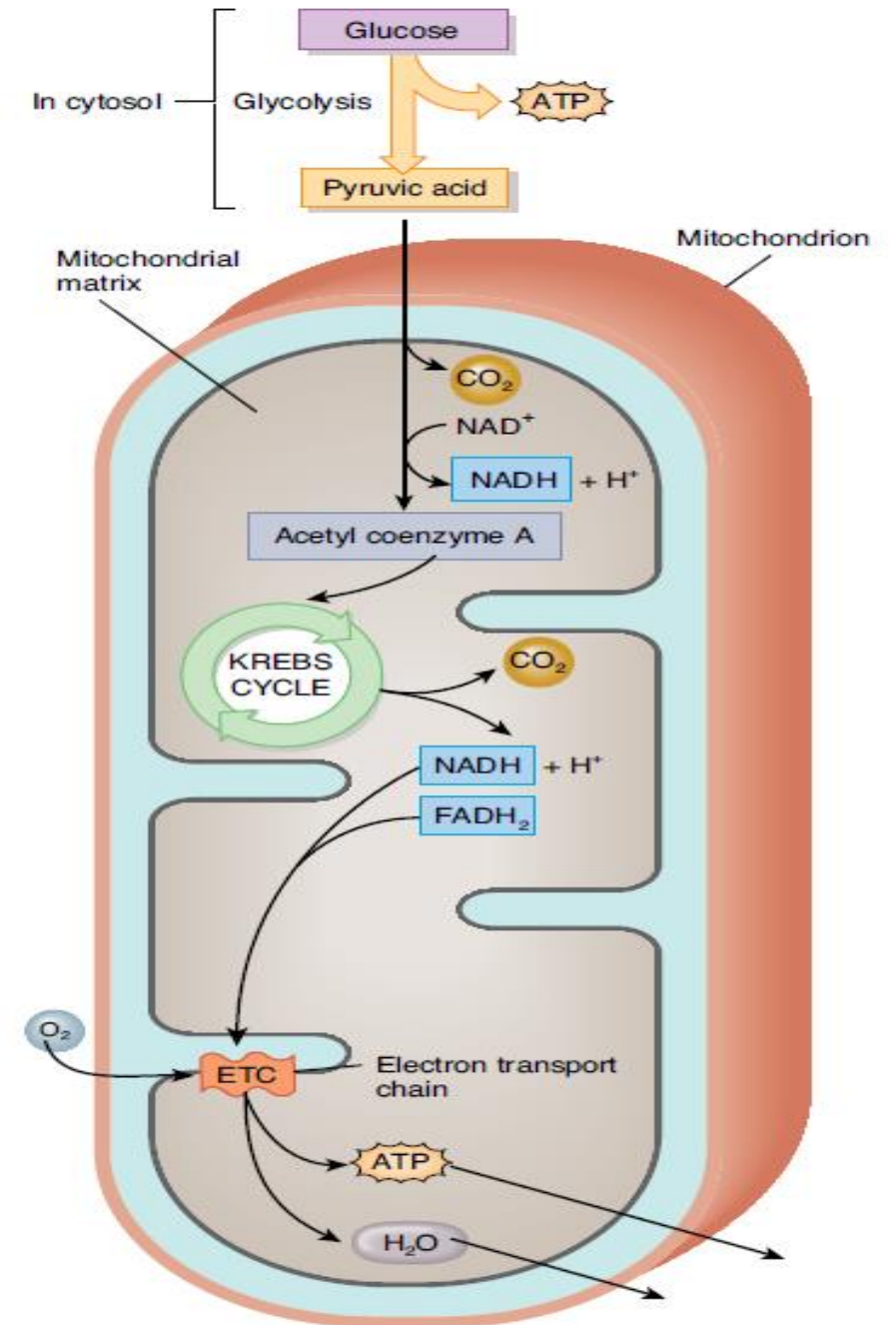
? Why is the production of reduced coenzymes important in the Krebs cycle?

- Because glycolysis does not require oxygen, it can occur under aerobic (with oxygen) or anaerobic (without oxygen) conditions.

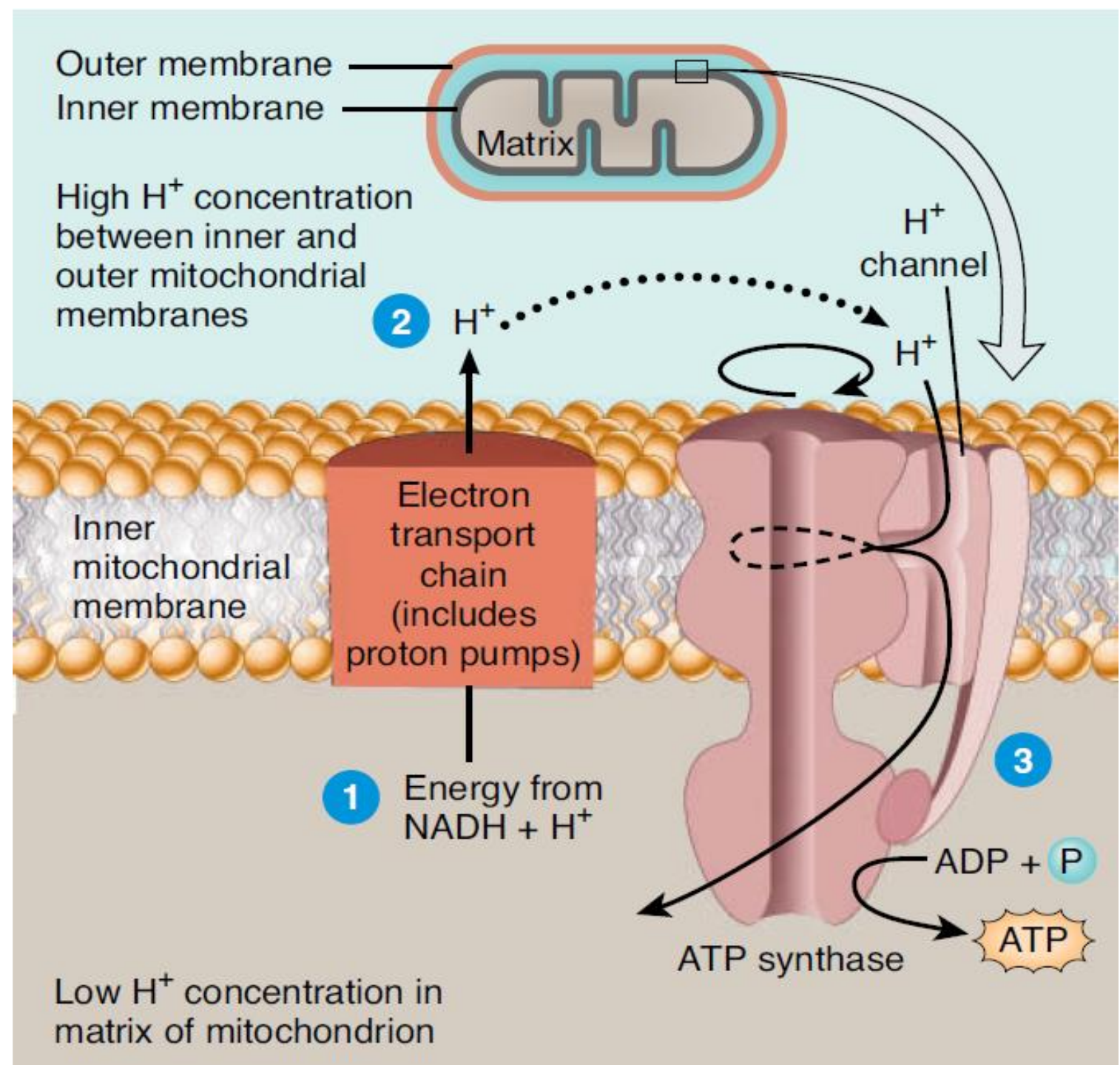
- By contrast, the reactions of the Krebs cycle and electron transport chain require oxygen and are collectively referred to as **aerobic respiration**.



Thus, when oxygen is present, all four phases occur: glycolysis, formation of acetyl coenzyme A, the Krebs cycle, and the electron transport chain.

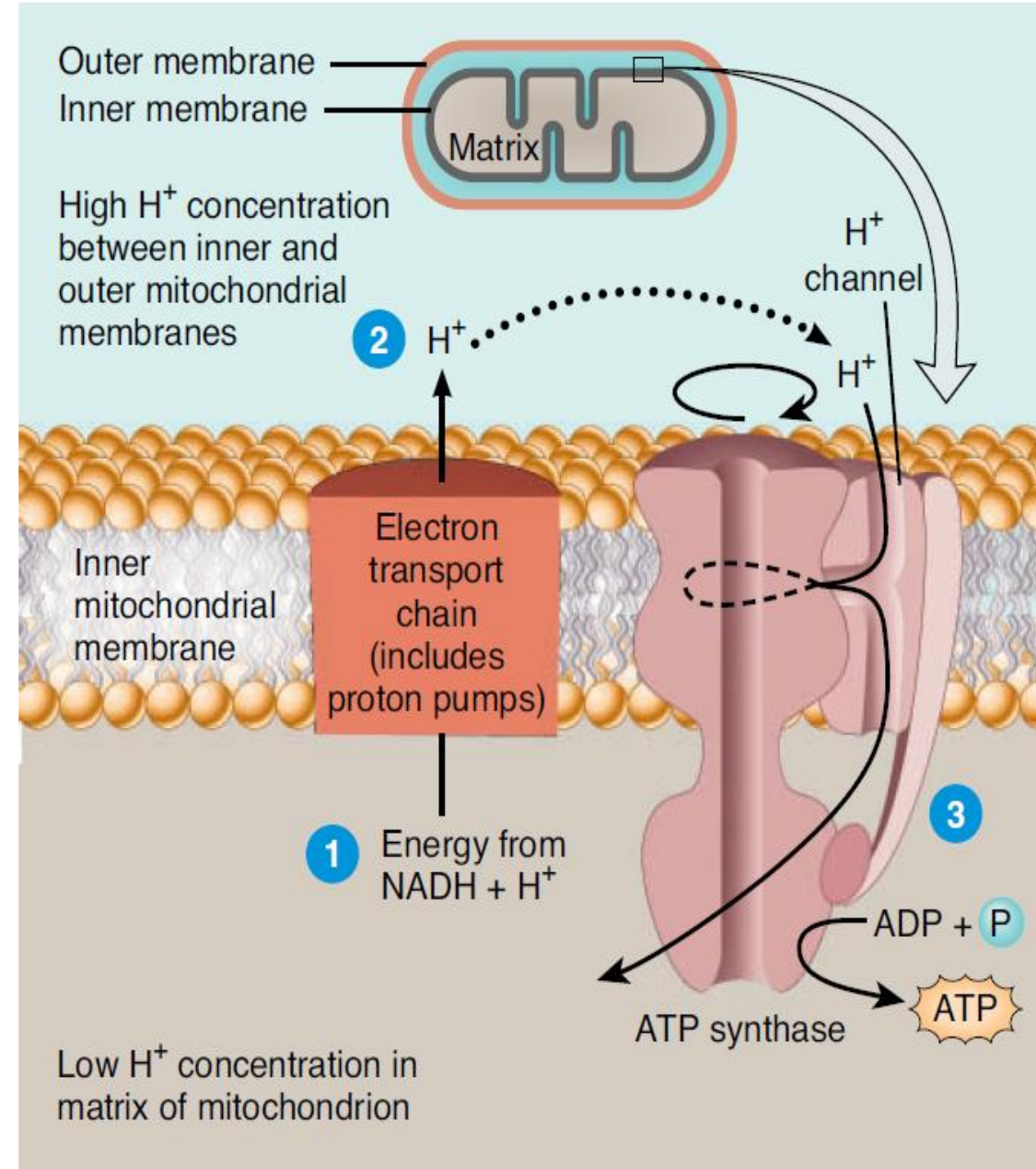


- Because this mechanism of ATP generation links chemical reactions (the passage of electrons along the transport chain) with the pumping of hydrogen ions, it is called **chemiosmosis**.



- The pumping of H produces both a concentration gradient of protons and an electrical gradient.
- The buildup of H makes one side of the inner mitochondrial membrane positively charged compared with the other side.
- The resulting electrochemical gradient has potential energy, called the **proton motive force**.
- Proton channels in the inner mitochondrial membrane allow H to flow back across the membrane, driven by the **proton motive force**. As H flow back, they generate ATP because the H channels also include an enzyme called ATP synthase.
- The enzyme uses the proton motive force to synthesize ATP from ADP and P (**ATP Synthase**).

The process of chemiosmosis is responsible for most of the ATP produced during cellular respiration.



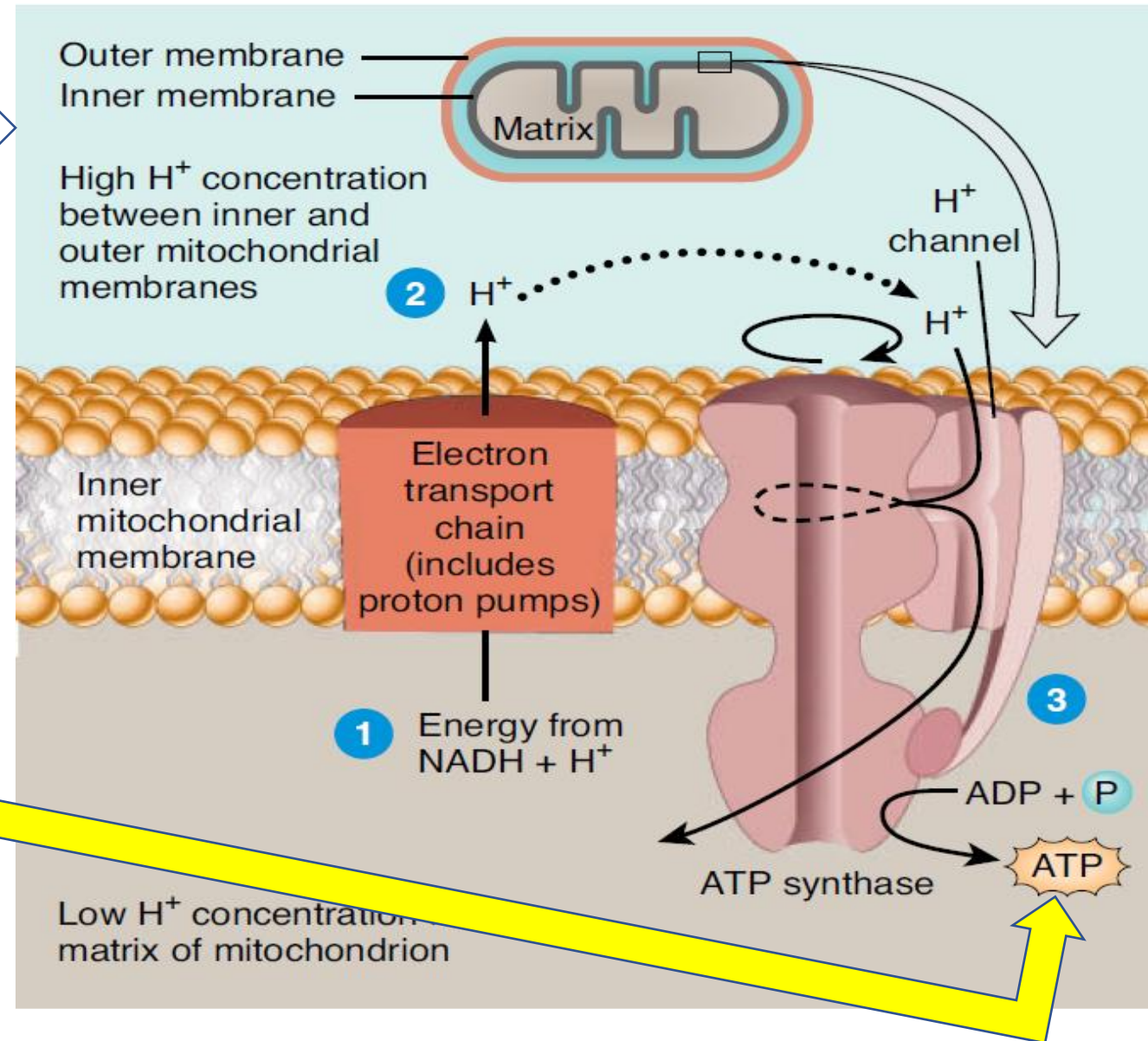
Electron transport chain

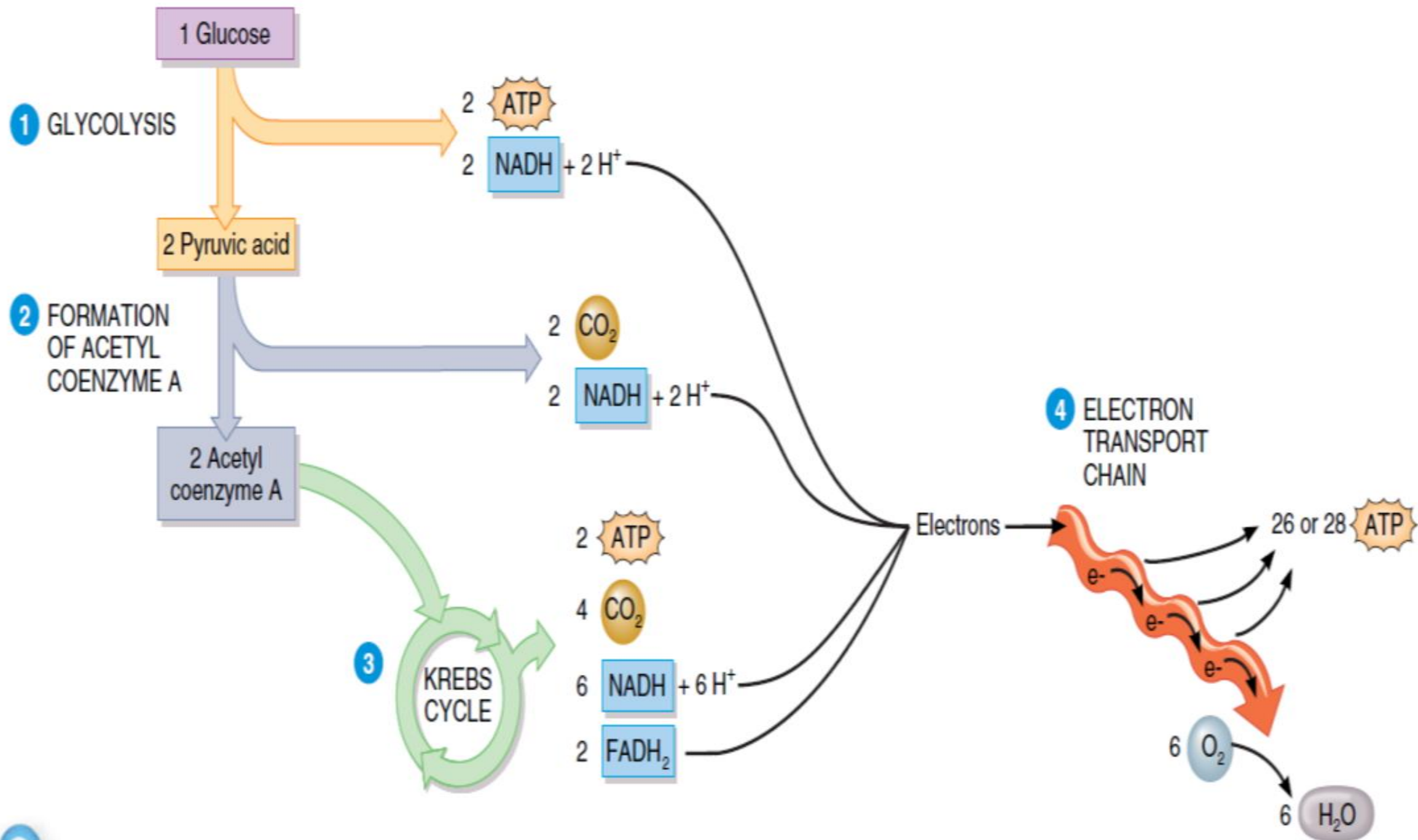
Plus 2 ATP (Glycolysis)
and 2 ATP (Krebs Cycle)

Total 30 or 32 ATP
from each Glucose
Molecule!

26 or 28

ATP





Glucose Anabolism

- Even though most of the glucose in the body is catabolized to generate ATP, glucose may take part in or be formed via several anabolic reactions.
- One is the **synthesis of glycogen**; another is the **synthesis of new glucose molecules from some of the products of protein and lipid breakdown**.

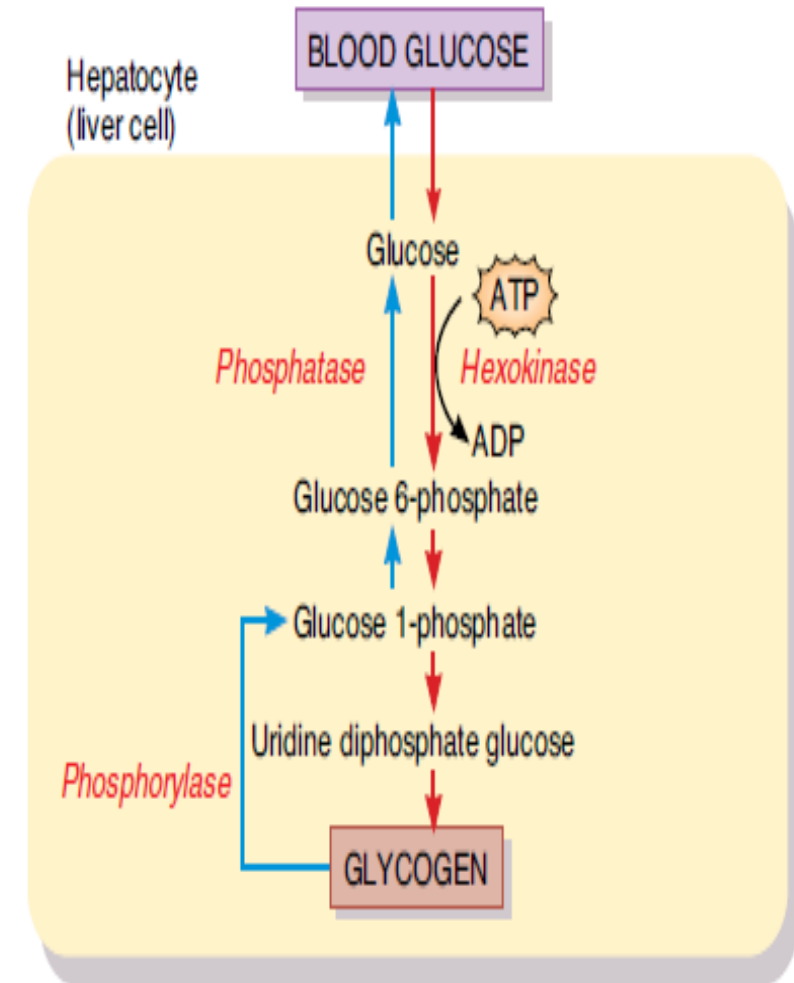
Glucose Storage: Glycogenesis

Glucose Release: Glycogenolysis

**Formation of Glucose from
Proteins and Fats:
Gluconeogenesis**

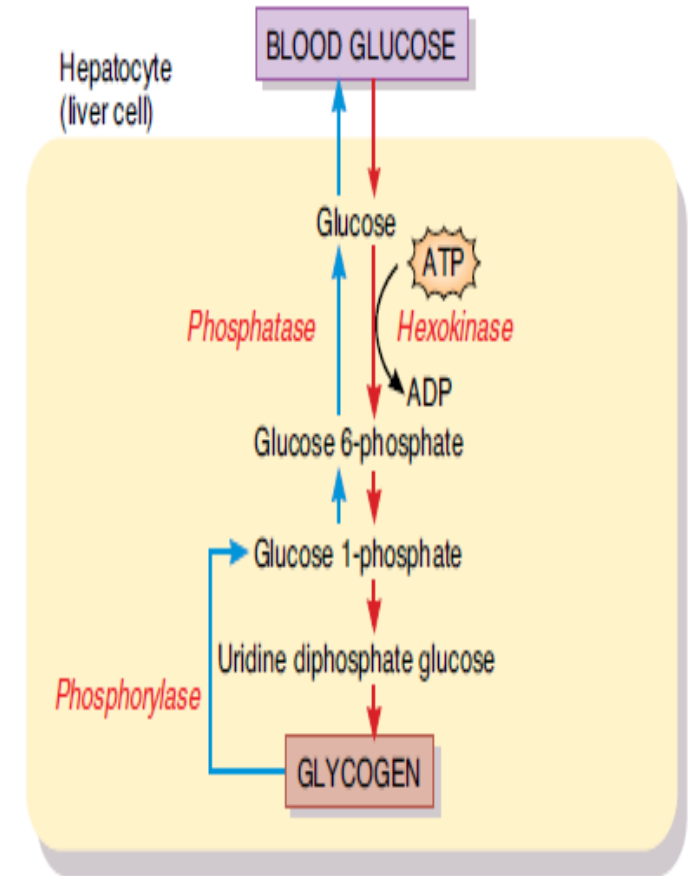
Glucose Storage: Glycogenesis

- If glucose is not needed immediately for ATP production, it combines with many other molecules of glucose to form glycogen.
- The hormone **insulin**, from **pancreatic beta cells**, stimulates hepatocytes and skeletal muscle cells to carry out **glycogenesis**, the synthesis of glycogen.
- The body can store roughly 75% in skeletal muscle fibers and the rest in liver cells.



Glucose Release: Glycogenolysis

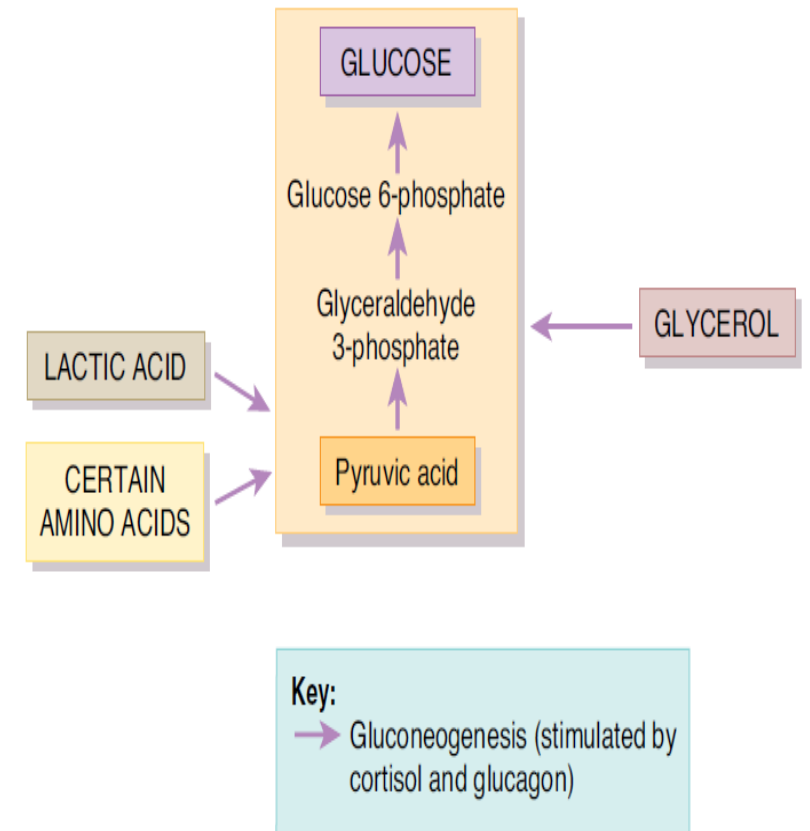
- When body activities require ATP, glycogen stored in hepatocytes is broken down into glucose.
- The process of splitting glycogen into glucose called **glycogenolysis**.
- Phosphorylase, is the enzyme that catalyzes glycogen, is activated by **glucagon** from pancreatic alpha cells and **epinephrine** from the adrenal medullae.



Formation of Glucose from Proteins and Fats: Gluconeogenesis:

- When your liver runs low on glycogen, it is time to eat.
- If you don't, your body starts catabolizing triglycerides (fats) and proteins.
- Large-scale triglyceride and protein catabolism does not happen unless you are starving, eating very few carbohydrates, or suffering from an endocrine disorder.
- The glycerol part of **triglycerides**, **lactic acid**, and certain **amino acids** can be converted in the liver to glucose. The process by which glucose is formed from these noncarbohydrate sources is called **gluconeogenesis**.

- Gluconeogenesis is stimulated by **cortisol**, and by **glucagon** from the pancreas.
- **Thyroid hormones** (T3,T4) mobilize proteins and may mobilize triglycerides from adipose tissue, making glycerol available for gluconeogenesis.





THANK
YOU